

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

M.Tech (Communication system Engg.) Sem - II University Examination, May-2011

EC 711 Principles of Wireless Networks

Date: 19.05.11, Thursday Time: 10:00 a.m to 01:00 p.m Maximum Marks: 70

- Instructions:** (1) All questions are compulsory.
(2) Indicate clearly, the option you attempt along with its respective question number.
(3) Figures to right indicate marks
(4) Rough work is done in the last page of main answer book, don't write anything on the question paper.

SECTION -I

Q-1 Answer the following

- (a) Draw the Architecture of PRNETs and explain in brief. 5
- (b) What is an Ad Hoc Network? Give its comparison with Wireless Sensor Networks. 5
- (c) Explain Wireless LANs in brief 2

- Q-2(a) List the routing protocol for wireless and wired network. Draw the TREE chart of Ad Hoc Routing Protocols. 5
- (b) Discuss Media Access in PRNETs. 5
 - (c) What is the difference between proactive routing and reactive routing? 3

OR

- Q-2(a) Give the comparison of IARP and IERP. Discuss in which scenario ZRP can be applicable? 5
- (b) Prove that an increase in sensor density by a factor of k improves the SNR by $10\log k$ db. 5
 - (c) How the problem of Count to infinity is resolved in DSDV? If new wireless node enters in to the network, how the routing table updated? 3

- Q-3(a) What is LEACH protocol? Discuss its importance for wireless sensor network. 5
- (b) Give comparison between AODV and DSDV protocol in detail. 5

OR

- Q-3(a) Draw the packet format for OLSR. Discuss the concept of message forwarding with respect to OLSR. 5
- (b) Justify the following statement with characteristic curve and necessary equation: 5
"The power advantage of using a multihop RF communication is more compared to single hop communication over distance X_d "

SECTION-II

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|---------------|--|-----------|
| Q-4(a) | Give the comparison of MANET and WSN with specific technical parameters in tabular format and justify the following statement with TRUE or FALSE with reson. "The nes-C and TinyOS must be used for an implementation of MANET access point" | 5 |
| (b) | How Video sensor network can useful for traffic and Security application? Explain with diagram | 5 |
| OR | | |
| Q-4(a) | Explain a Unicast Geographic Routing with six nodes greedy forwarding strategy. | 5 |
| (b) | What is OLSR? How it differs from OSPF. Discuss the advantage and disadvantages of OLSR | 5 |
| Q-5(a) | What is S-MAC protocol? How it different than normal MAC protocol? | 5 |
| (b) | Discuss the role of Sensor Network Platforms in real time product development. | 5 |
| (c) | Justify the statement :
"DSR is reactive protocol which has lower PDR and lower average end-to-end delay". | 3 |
| OR | | |
| Q-5(a) | Explain Route Discovery and Maintenance of DSR Protocol | 5 |
| (b) | What is Geographic, energy aware routing? Explain with necessary figures | 5 |
| (c) | Give the structure of routing header | 3 |
| Q-6 | Write a short note on (Any Three) | 12 |
| (a) | Dynamic Source Routing (DSR) | |
| (b) | Ad Hoc On-Demand Distance Vector Routing (AODV) | |
| (c) | Sensor network programming challenges | |
| (d) | Clustering in Sensor Network | |
| (e) | Cluster Switch Gateway Routing (CSGR) | |